

TEOREMA DI BOLZANO WEIERSTRASS

SIA $f(x)$ CONTINUA IN $[a, b]$. ALLORA $f(x)$ ASSUME UN VALORE DI MASSIMO E MINIMO ASSOLUTO IN $I[a, b]$ OUVERO

$$\exists x_1, x_2 \in [a, b]: f(x) \leq f(x_1) \leq f(x_2)$$

$$m = f(x_1) = \text{minimo di } f(x) \text{ in } [a, b]$$

$$M = f(x_2) = \text{MASSIMO di } f(x) \text{ in } [a, b]$$

ESERCIZIO n°1

$$f(x) = 4x^3 - 5x^2 + 2x - 3$$

$$f'(x) = 12x^2 - 10x + 2$$

$$x_{1,2} = \frac{10 \pm \sqrt{100 - 4(12)(2)}}{2 \cdot 12} = \frac{10 \pm \sqrt{100 - 96}}{24} =$$

$$\frac{10+2}{24} = \frac{1}{2}$$

$$\frac{10-2}{24} = \frac{1}{3}$$

$$f(x) \geq 0 \quad 12x^2 - 10x + 2 \geq 0$$

$$x_{1,2} = \frac{10 \pm \sqrt{100 - 4(12)(2)}}{2 \cdot 12} = \frac{10 \pm \sqrt{100 - 96}}{24} =$$

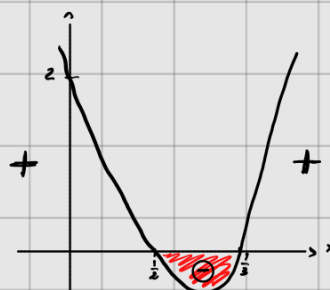
$$\frac{10+2}{24} = \frac{1}{2}$$

$$\frac{10-2}{24} = \frac{1}{3}$$

$$x_1 \geq \frac{1}{2}$$

$$x_2 \geq \frac{1}{3}$$

	$\frac{1}{2}$	$\frac{1}{3}$	
—			—
—	+	+	—
—	—	+	—
—	—	+	—
+	—	+	



$$x < \frac{1}{2} \vee x > \frac{1}{3} \rightarrow f(x) > 0 \quad \frac{1}{2} < x < \frac{1}{3} \rightarrow f(x) < 0$$

ESERCIZIO n°2